Modelling and Simulation of Flash Separator and Distillation Column in Aspen HYSYS: Project Report

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1. Problem Statement and Objective:

Using Aspen HYSYS, model and simulate a Flash Separator, which runs below the feed's bubble point, connected to a Distillation column to effectively separate and extract Ethanol from a feed composed of Ethanol and water (azeotropic mixture), given the initial process information, aiming for a recovery below 89% (azeotropic limit for normal distillation) through the setup. Also, to make a sensitivity Case study to see how different inlets affect outlet flow and composition

Feed flow	100 kmol/h
Feed composition	40 mol% Ethanol 60 mol% Water
Feed Temperature	25 °C
Feed Pressure	1 atm (101.325 kPa)
Flash Pressure	1 atm
Condenser Pressure	101.3 kPa
Reboiler Pressure	105 kPa
Reflux ratio	2.5

Distillate Rate	35 kmol/h

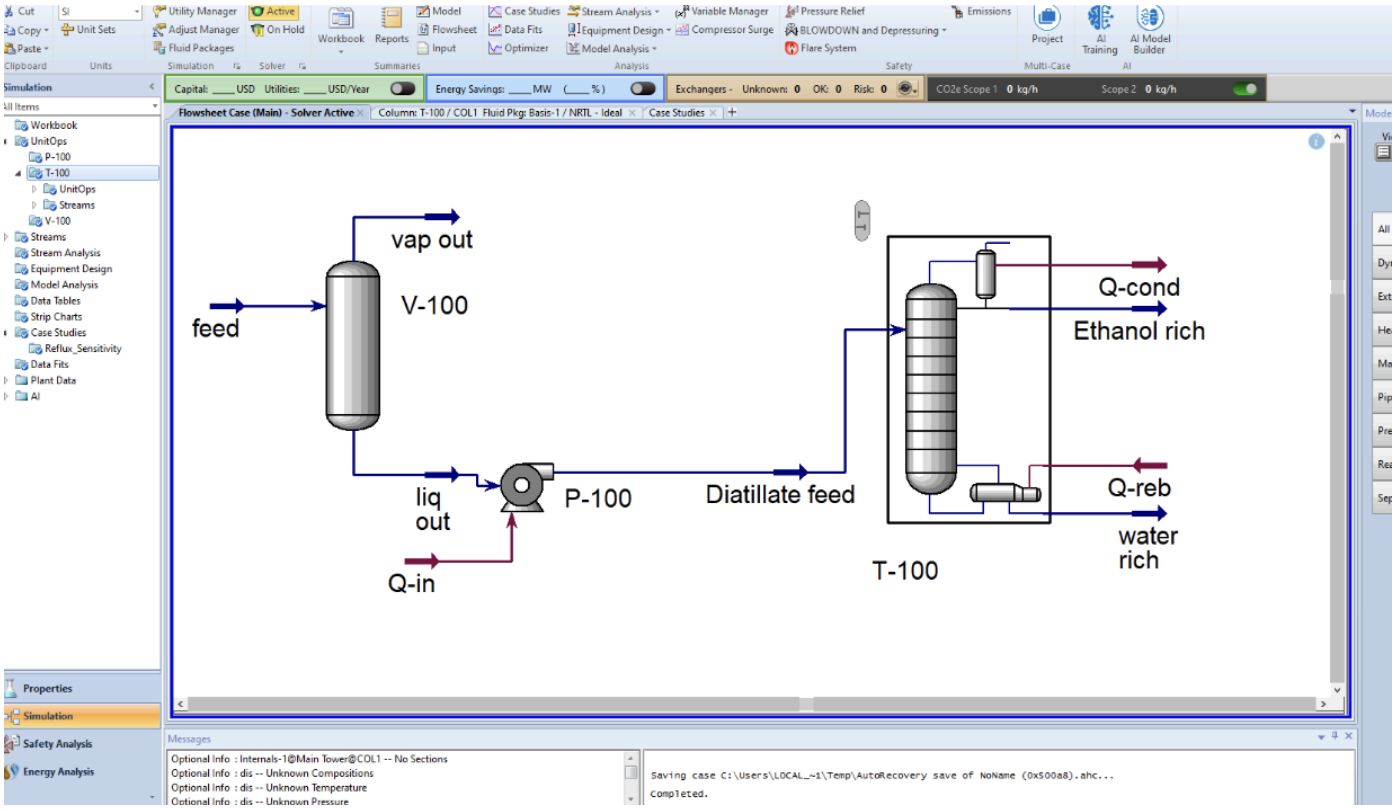
2. Assumptions:

- Steady-state operation
- Vapour–liquid equilibrium is reached on every stage of the column (equilibrium-stage model, 100% stage efficiency)
- The flash separator is an ideal single equilibrium stage (isothermal)
- The flash separator operates at 25 °C and 101.3 kPa, below the bubble point of the feed
- Flash calculations outside HYSYS use Van Laar activity coefficients and Antoine vapour pressures
- Neglecting any energy exchanges or changes
- Total condenser and thermosiphon reboiler
- No chemical reaction
- Thermodynamic behaviour is described by the NRTL model

2. Process description and Flowsheet:

According to our problem statement, the process consists of a flash separator and a distillation column. I first set up the separator and successfully created two outlets: one liquid-rich and the other vapour-rich, but in a practical sense, because it runs below the feed's bubble point it acts as a feed drum hence protecting it from vapour. As a result, there was no vapour outlet. So, the feed comes out of the liquid stream identical to how it went in, and now this stream is connected to the distillation column with 5 stages (default in Aspen). For the column to operate, the condenser and reboiler have a pressure difference, so a

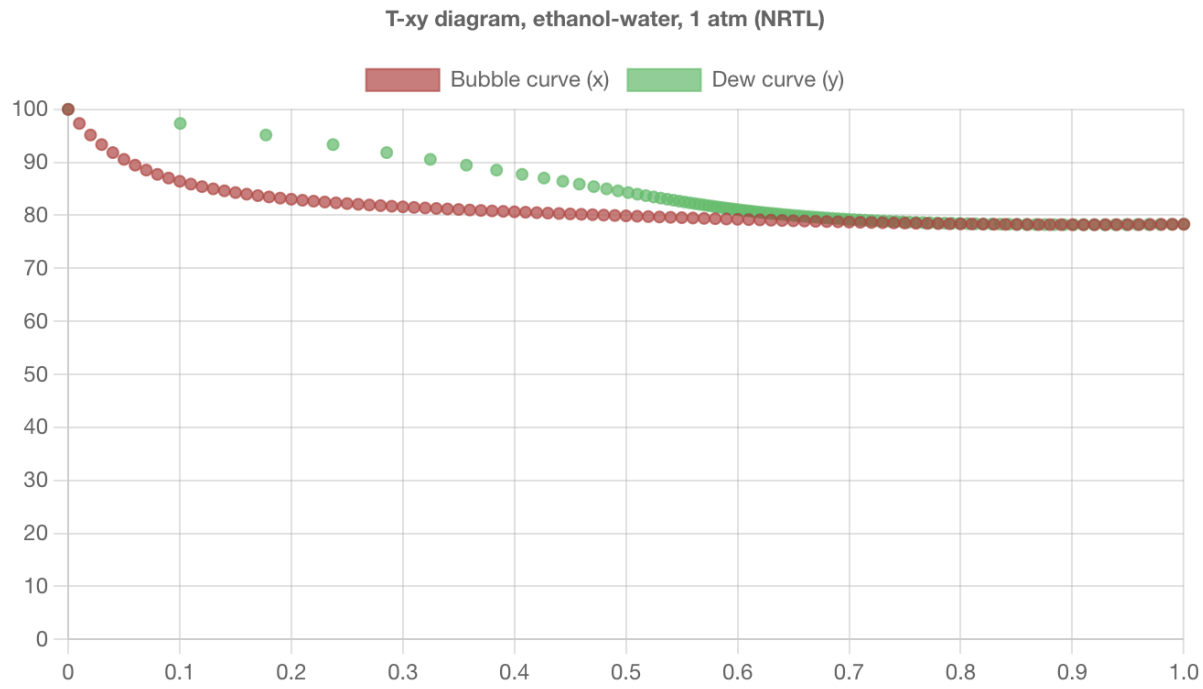
pump has been added to raise the distillate feed pressure, and after specifying distillation specifications, the flowsheet was completed.



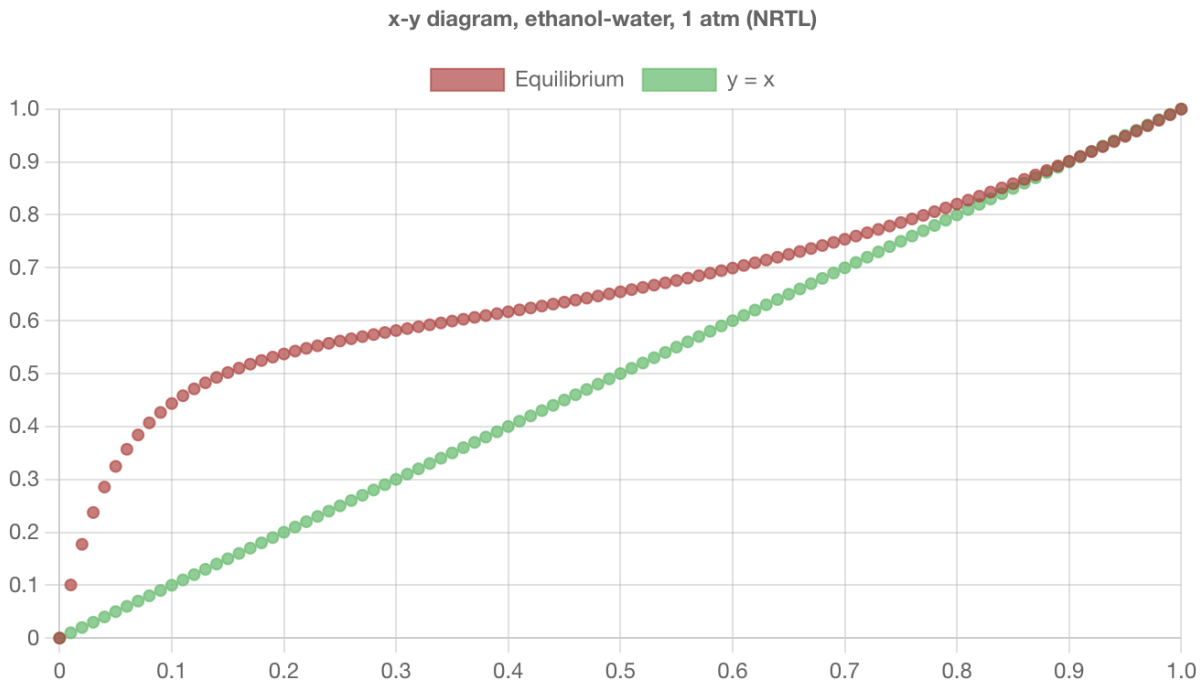
4. Simulation setup:

- Before setup, we are required to select components and the property package that is the best fit for simulating any process involving the components. This step is usually well laid out, and Aspen will raise warnings for packages that may not be appropriate. I have concluded that NRTL is the best choice based on recommendations from Aspen, because the mixture is a non-ideal azeotropic mixture; NRTL fits polar, hydrogen-bonding mixtures well, has a non-randomness parameter (alpha), and handles the azeotrope and the temperature dependence.
- The Aij, Bij, and alpha parameters for this setup from Aspen are -109.634, 1332.31 and 0.303. Since the version I use doesn't have a display of T-x-y and x-y tables, I have plotted the data in Excel

and extracted the charts. These are used to confirm whether the azeotrope appears near 89 mol% and around 78°C.



Also, in this graph, the lower curve gives the bubble point, and the upper curve gives the dew point



- For the Setup to work for a successful simulation, the data parameters given in the problem statement need to be entered in the Parameters and Composition tab for the separator first, then the distillate column.
- After the separator gives a converged simulation, we can go to the next step: The pump. It becomes a necessity here, as we need to deal with pressure differences between the column and separator. The pump requires a specification of delta pressure (the difference between initial and final pressures), which is 3.7
- The pumped feed now becomes the distillation feed for the column; here we use a 5-stage distillation tower again, as recommended by Aspen. Then we put in the parameters: composition and flow rate remain the same as the outlet from the separator, and 2 main specifications to make this column include reflux ratio and distillate feed (many specifications can be used, but these are basic and a minimum mandate of 2 specs)

5 Theory and model :

This is the theory that Aspen models systems after, based on parametric requirements

- Flash separation: At equilibrium, $y_i = K_i x_i$, where K_i is the equilibrium ratio of component i. With feed fraction z_i and vapour fraction $\Psi = V/F$, the Rachford–Rice equation is $\Sigma\, z_i(K_i - 1) / [1 + \Psi(K_i - 1)] = 0$, and $x_i = z_i / [1 + \Psi(K_i - 1)]$. [One sentence on how the flash temperature affects Ψ and the vapour composition.]
- Distillation column: The column is modelled stage by stage with the MESH equations (material balances, equilibrium, summation and enthalpy balances). The relative volatility is α_i□ = K_i / K□, and the separation becomes easier as α moves away from 1. For a shortcut check, the Fenske equation gives the minimum number of stages at total reflux, $N_{min} = \ln[(x_D/(1 - x_D))((1 - x_B)/x_B)] / \ln \alpha$, and the Underwood equations give the minimum reflux ratio R_min. [State which one you used.]
- Overall balances: Total: $F = D + B$. Component: $Fz = Dx_D + Bx_B$. Recovery of the light key in the distillate: $Dx_D / (Fz)$. Energy: $Q_{reboiler} - Q_{condenser} = D \cdot h_D + B \cdot h_B - F \cdot h_F$.

6. Hand Calculations and Validation :

➢ Overall material balance:

Stream	kmol/h	kg/h	Ethanol (mole fraction)	Ethanol (kmol/h)	Ethanol (wt%)
Distillate feed (in)	100.00	2924	0.4001	40.011	63.0
Ethanol rich (distillate)	44.30	1780	0.7901	35.000	90.6
Water rich (bottoms)	55.70	1143	0.0893	4.975	20.1

Residual (in − out)	0.000	1	n/a	0.036	n/a

- Total balance**:** 100.00 = 44.30 + 55.70 kmol/h, and the mass residual is 1 kg/h (0.03%), which is rounding in the displayed flows.
- Ethanol balance**:** the residual is 0.036 kmol/h (0.09% of feed ethanol).
- Conclusion**:** both balances close. The process also satisfies the water balance (59.99 in, 9.30 + 50.73 out).
- Product quality**:** the distillate is about 79 mol% (90.6 wt%) ethanol, below the azeotrope at roughly 89 mol%, so it is reachable by ordinary distillation.

➢ Flash Check:

HYSYS gives a vapour fraction of 0 and a vapour-out flow of 0. A hand check confirms this: with $K = \gamma P_{sat}/P$, the sum of $z_i K_i$ is 0.05 (ideal) to 0.12 ($\gamma = 2.5$), well below 1, so the feed is entirely liquid and $\Psi = 0$. The feed bubble point is calculated to be 80.7 °C, about 56 °C above the operating temperature.

Check	**Calculated**	**HYSYS**
Distillate bubble point (x = 0.790, 101.3 kPa)	78.40 °C	78.27 °C
Bottoms bubble point (x = 0.0893, 105 kPa)	88.16 °C	87.97 °C

➢ Short-cut column check:

Using $x_D = 0.790$ and $x_B = 0.0893$, the separation factor is $S = (0.790/0.210)(0.9107/0.0893) = 38.4$, so $\ln S = 3.65$.

Quantity	Hand calculation	HYSYS
α (geometric mean, trays 1–11)	2.01	n/a
N_min (Fenske)	5.2 stages	n/a
Actual equilibrium stages	n/a	11 (10 trays + reboiler)
N_min / N	0.47	n/a

A ratio close to 0.5 is typical of a column running at about 1.2–1.5 times the minimum reflux, so the stage count is reasonable. α comes from your K-value plot and varies from about 1.2 at the top to about 7.8 at the reboiler, so treat 5.2 as an estimate. The top/bottom average (α = 2.99, N_min = 3.3) overstates the reboiler end and is less reliable.

7. Results:

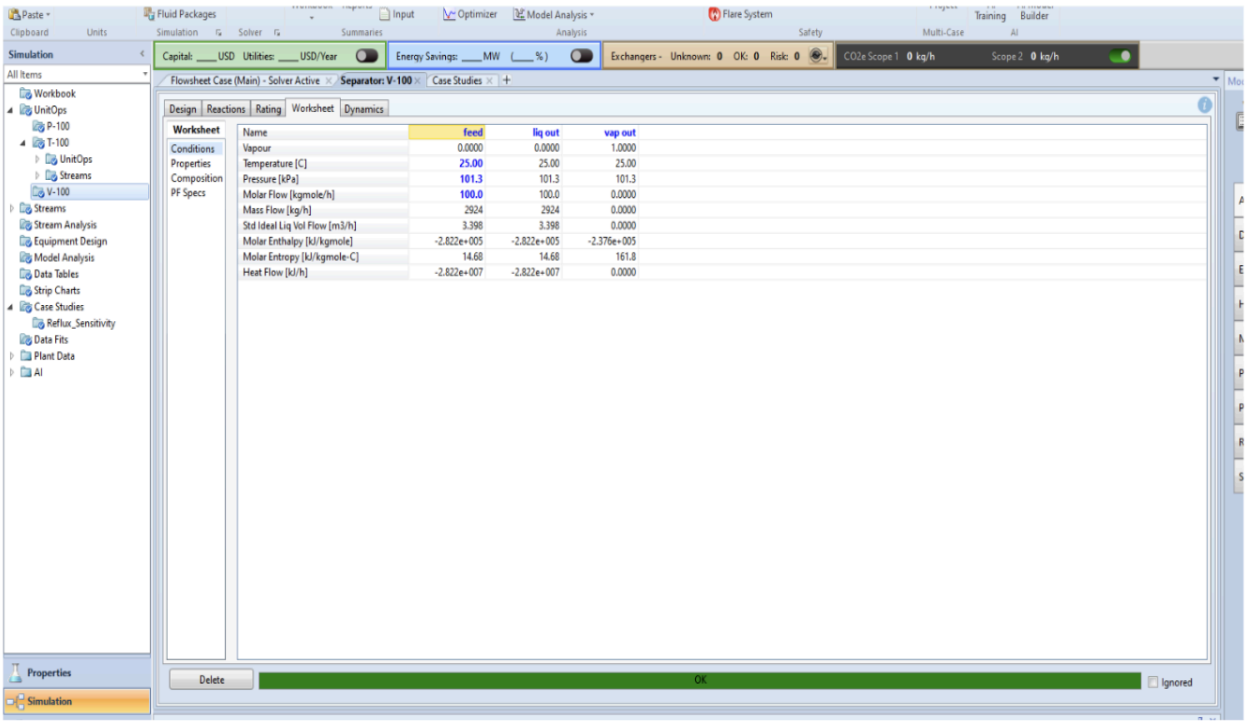
➢ Convergence:

The simulation converged with no errors and no warnings for the the V-100 Flash Seperator but for the the distillate column it took a while for it converge as distillate rate was found out using trial and error based on 2.5 reflux ratio.

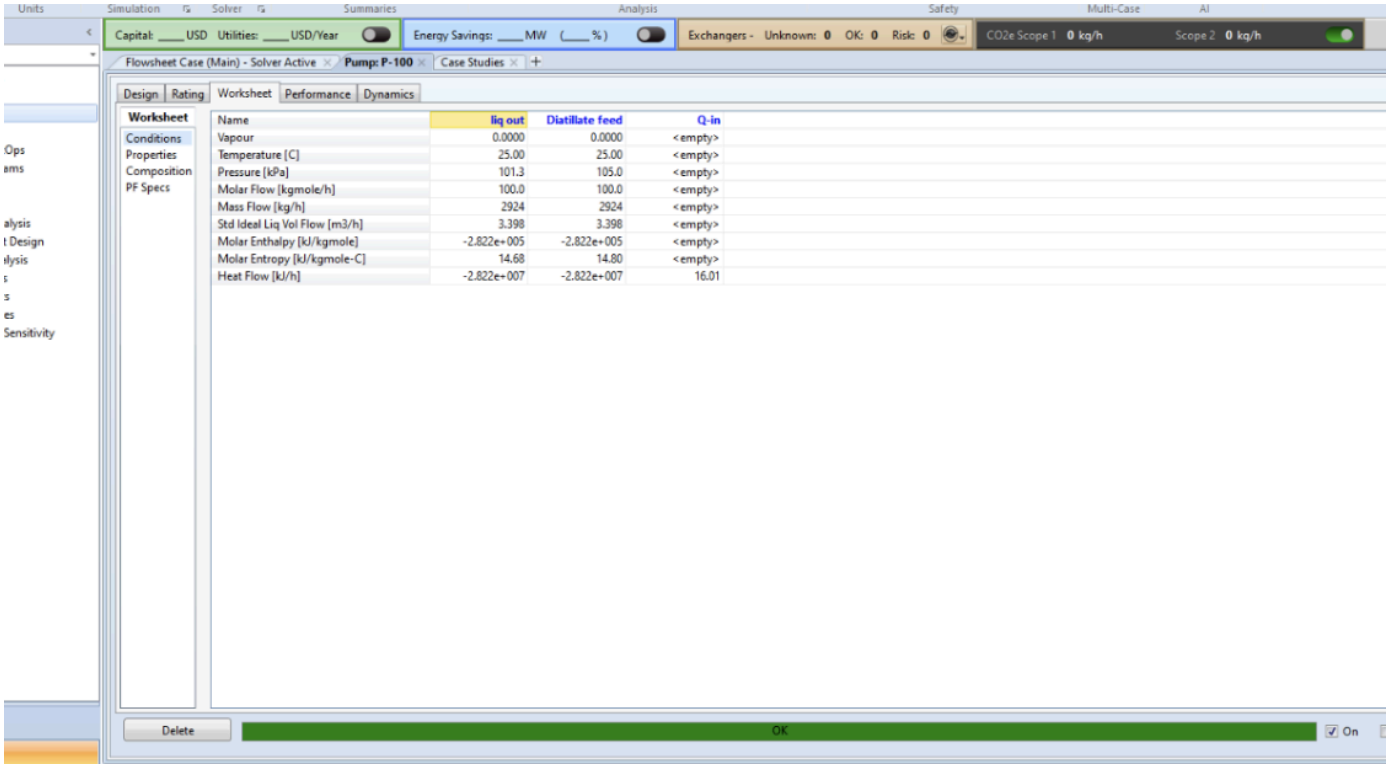
➢ Stream Results:

1. Flash Seperator V-100 : Here the flash separator does no real separation as the it runs below the feed's bubble

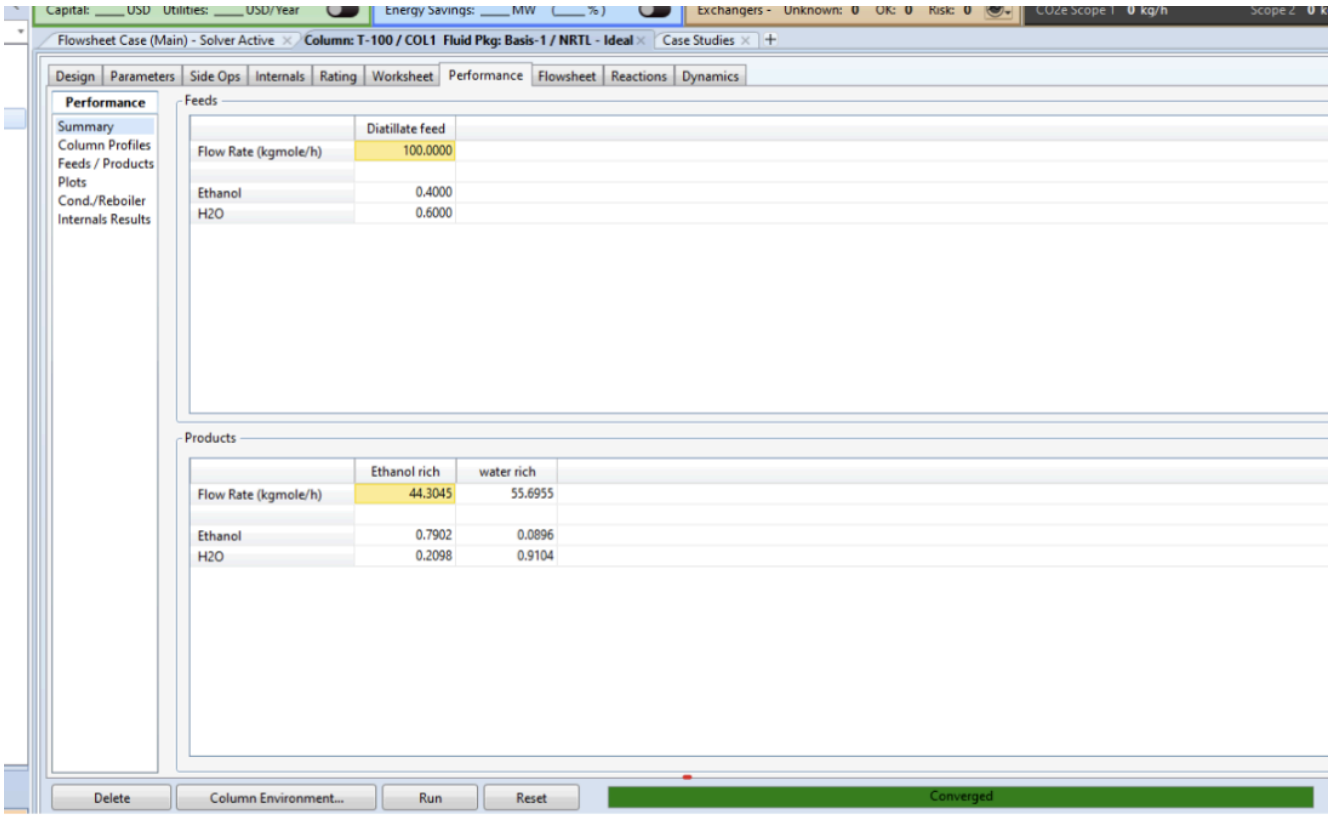
point acting mainly as a feed drum so vapour outlet is zero but liquid outlet is the same as inlet

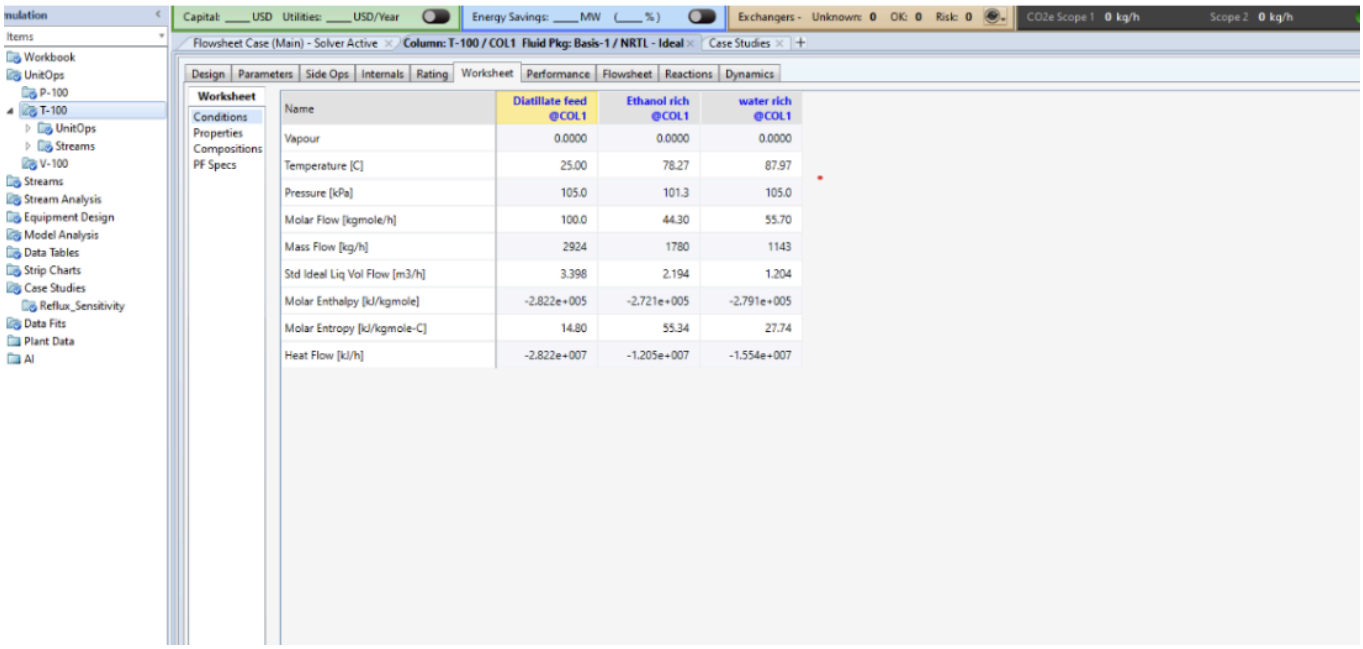


2. The pump:Here as the flash separator is merely the feed the work to increase the feed pressure for distillate column is done by the pump by applying sufficient energy and work



3. The distillation column:The distillation column provides the final outlet composition and flow rates and how much ethanol has been recovered the composition change happens through various stages of the column

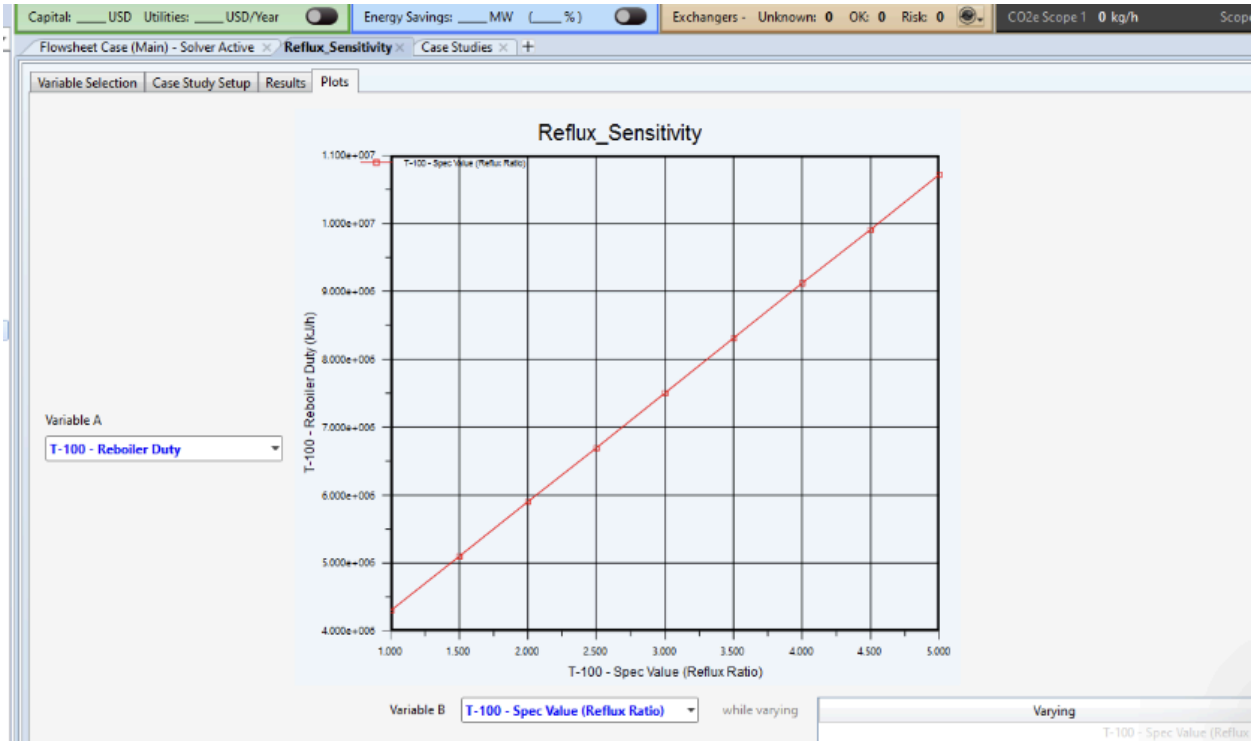




8. Sensitivity Case Study:

The case study feature has been used sensitivity analysis with reflux ratio as constant and see how distillate molar flow and reboiler duty vary

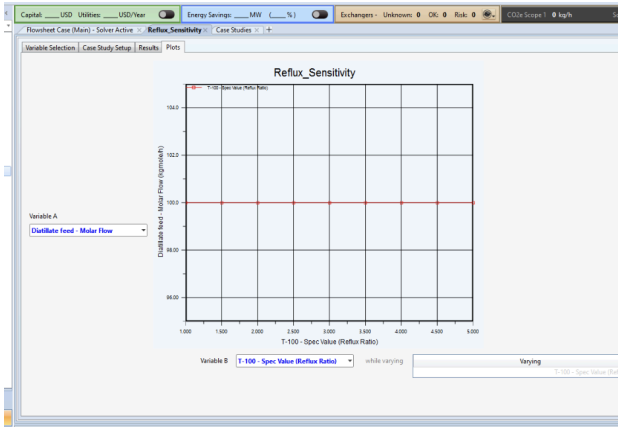
with it



The reboiler duty rises almost linearly with reflux ratio, from about 4.3 × 10^6 kJ/h (1.2 MW) at a reflux ratio of 1.0 to about 1.07 × 10^7 kJ/h (3.0 MW) at 5.0, an increase of about 150%. Each additional unit of reflux ratio costs roughly 1.6 × 10^6 kJ/h (0.44 MW). At a reflux ratio of 2.5 the duty is about 6.7 × 10^6 kJ/h (1.9 MW). The linear trend is consistent with a fixed distillate rate: the boil-up V = D(R + 1) grows linearly with the reflux ratio R, and the reboiler duty is roughly the boil-up multiplied by the latent heat.

This study recorded only energy, so the effect of reflux ratio on distillate purity and ethanol recovery was not recorded and the point of diminishing returns cannot be located from these results. In general the purity gain flattens as the reflux ratio rises while the energy cost keeps growing linearly, and ethanol–water cannot exceed the azeotrope at about 89 mol% ethanol however much reflux is used. The minimum reflux ratio was not found directly. The Fenske result (N_min/N = 0.47) suggests the column is running at a moderate multiple of the minimum

reflux, as is typical of a column operating at about 1.2 to 1.5 times R_min.
The flow rates are linear with reflux ratio as the feed was help constant



9. Inference :

The flash separator V-100 operates at 25 °C and 101.3 kPa, below the calculated bubble point of 80.7 °C for the 40 mol% ethanol feed, so the feed stays entirely liquid (vapour fraction 0) and the vessel performs no separation; it acts as a feed drum ahead of pump P-100. A calculation outside HYSYS shows it would only separate the feed between 80.7 and 88.0 °C, giving, for example, a vapour fraction of 0.57 and a vapour of 54 mol% ethanol at 83 °C. All the separation in this design is therefore done by the column. The column splits the 100 kmol/h feed into 44.30 kmol/h of distillate at 79.0 mol% (90.6 wt%) ethanol and 55.70 kmol/h of bottoms at 8.9 mol% (20.1 wt%) ethanol, which corresponds to an ethanol recovery of 87.5%, with 12.4% of the feed ethanol lost in the bottoms. The hand calculations agree with HYSYS: the overall mass balance closes to 0.03% and the ethanol balance to 0.09%, a calculated VLE model reproduces the distillate and bottoms bubble points to within 0.2 °C, and the Fenske minimum of 5.2 stages compares with 11 actual stages (N_min/N = 0.47). The reflux ratio study shows that the reboiler

duty rises linearly from 4.3×10^6 kJ/h at a reflux ratio of 1 to 1.07×10^7 kJ/h at 5, about 0.44 MW for each unit of reflux ratio. The base case, at a reflux ratio of about [2.5], needs roughly 6.7×10^6 kJ/h (1.9 MW). Because the purity response to reflux ratio was not recorded, the point of diminishing returns cannot be quantified from this study. I would keep the base-case reflux ratio as the operating point and, before raising it to recover more of the 12.4% of ethanol lost in the bottoms, measure purity and recovery against reflux ratio and consider adding stages instead, since extra reflux costs energy continuously.

10. Real-World Implications :

The reboiler duty is the main operating cost of a distillation column, and this study shows what it costs to push the separation harder: at the base case the column uses about 3.8 MJ of reboiler heat per kg of distillate (6.7×10^6 kJ/h for 1780 kg/h), and raising the reflux ratio from 2.5 to 4.0 would increase it by about 36% (to about 9.1×10^6 kJ/h). Because ethanol–water has an azeotrope at about 89 mol% (95.6 wt%) ethanol, no amount of extra reflux can produce anhydrous ethanol, so higher purities need a different technique such as an entrainer or molecular sieves. The feed enters at 25 °C, so part of the reboiler duty simply heats the cold feed, and there is a clear opportunity for heat integration: preheating the feed with the hot bottoms stream (88 °C) would cut the reboiler duty, and it would also allow the flash vessel to do useful separation if it were run in the 80.7 to 88.0 °C range. A real column would differ from this equilibrium-stage model in several ways: tray efficiencies below 100% would need more than 10 trays, pressure drop would raise the reboiler temperature, and the higher vapour rates at high reflux (the boil-up at a reflux ratio of 5 is about 1.7 times that at 2.5) could cause flooding and need a wider column. The NRTL property method is a sound choice for ethanol–water, but the results depend on its

binary parameters, and a different method would shift the K-values and therefore the stage requirement.

11. Limitations :

- The equilibrium-stage model assumes 100% stage efficiency, so a real column needs more trays
- Results depend on the property method and its binary parameters
- Hydraulics (column diameter, flooding, pressure drop) were not designed
- No heat integration or economic optimisation was done
- Mostly importantly the flash separator was not heated causing it to act as a feed drum
- The reflux ratio study recorded only reboiler duty, so the purity and recovery response was not measured and the optimum reflux ratio was not determined.